10

Blood Flow Dynamics

We are now going to continue our journey by exploring the physiology of extracorporeal membrane oxygenation (ECMO) support. You may find the chapters to follow to be the cornerstone of the understanding of ECMO. It is a common adage to hear the following:

"ECMO is easy to initiate, the challenge is in the management."

I fully agree with this sentiment, but the challenge in management is rarely what we think. The stalwarts of good critical care support remain consistent – medications, antibiotics, nutrition, and rehabilitation. However, the true challenge in managing patients on ECMO involves developing an appreciation, understanding, and familiarity with the physiology of extracorporeal support – the following chapters will be aimed to equip you with the tools needed to do just that.

Let's start by developing our understanding of the limits of blood flow and the ECMO circuit.

WHY IS BLOOD FLOW SO ESSENTIAL IN ECMO SUPPORT?

You will recall that there are two primary parameters that can be manipulated in ECMO support: **sweep gas flow**, which contributes to ventilation/CO_2 removal, and **blood flow**, which primarily contributes to oxygenation.

Why is this the case? Shouldn't support just be support? Shouldn't a higher oxygenated gas flow just increase oxygenation?

To answer this, we will have to return to our first chapter on the physiology of oxygen delivery.

Remember when it really comes to oxygen *delivery*, the ultimate determinant is hemoglobin, because hemoglobin is just so much better at delivering oxygen due to its ability to bind oxygen with increasing affinity in the heart lungs as well as the ability to change forms to and dump oxygen in the periphery (Fig. 10.1).

We covered in great detail what this means for the body in terms of delivering oxygen – that the better we can leverage the efficiency of hemoglobin, through optimizing saturation/hemoglobin/cardiac output, the better we can deliver oxygen.

Now let's apply this concept in the context of the ECMO circuit. You will recall for delivery of oxygen in the body we have the following equation:

$$DO_2 = 1.34 \times SaO_2 \times Hb \times CO + PaO_2 \times 0.003$$

which can be simplified to the following relationship:

$$DO_2 \rightarrow SaO_2 \times Hb \times CO$$

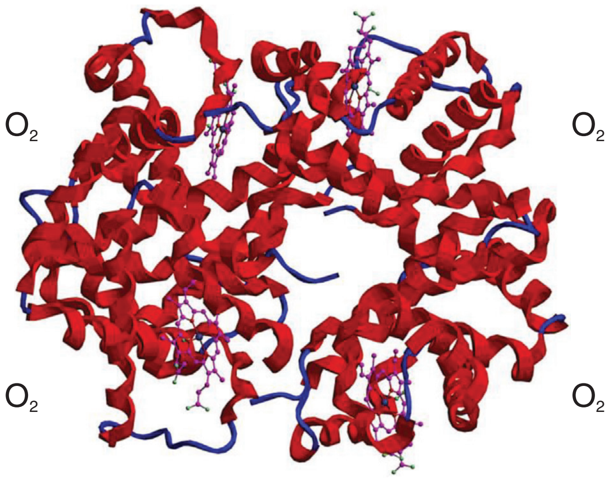

FIG. 10.1 Hemoglobin as a means of delivering oxygen. (From Raimundo79/Shutterstock.com.)

For the delivery of oxygen by the ECMO circuit, the determinants become oxygen saturation, hemoglobin, and instead of cardiac output, the blood flow through the ECMO circuit (Fig. 10.2).

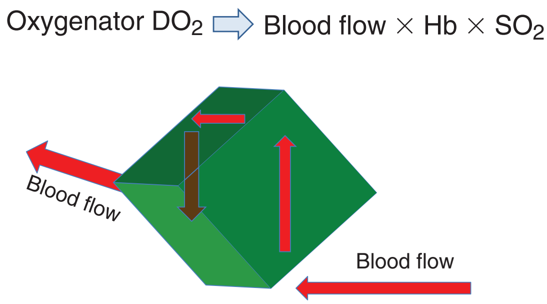

FIG. 10.2 Oxygen delivery capability of the oxygenator

WHAT ABOUT FiO_2?

Doesn't increasing FiO_2 improve the delivery of oxygen to the ECMO circuit much the same way that increasing FiO_2 administered to the lungs can increase oxygen delivery? Yes and no. The FiO_2 dial on the blender alters the amount of oxygen running through the oxygenator and drives the gradient at the membrane level, but ultimately, it does not matter how much oxygen diffuses across (otherwise known as your PaO_2); what ultimately matters to the delivery of oxygen is the saturation of hemoglobin by that oxygen.

At a certain point, it does not matter how high the PaO_2 of the blood coming out of the oxygenator is, there is a maximum saturation after which hemoglobin does not get much more saturated and the hemoglobin saturation curve flattens out (Fig. 10.3).

The oxygenator is very good at saturating blood, and there is rarely a problem with maintaining adequate saturation of blood coming out of the oxygenator. Rather at this point, we can start to appreciate that the determinants of oxygen delivery of the oxygenator are saturation of the

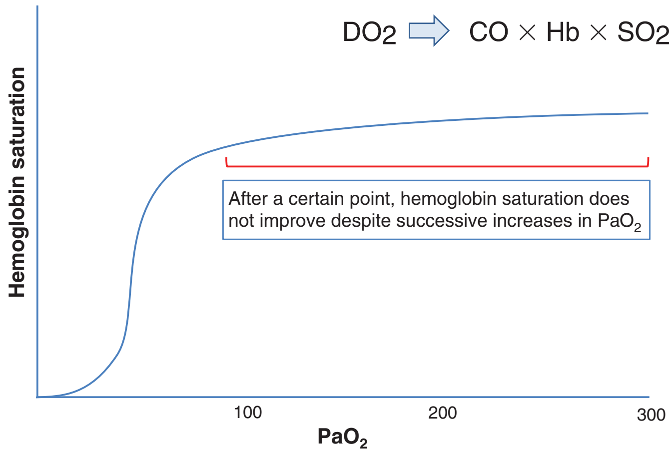

FIG. 10.3 Oxygen hemoglobin dissociation curve

blood coming out of the oxygenator (largely expected to be 100%), hemoglobin concentration, and, ultimately, the blood flow coming out of the oxygenator to the extent that we are able to provide this flow.

So more blood flow, better oxygen delivery – sounds easy enough. If you need more oxygen delivery, you provide more blood flow, right? As we will see, the ability to provide flow has been limited both by the pump/circuit and by the body.

WHAT ARE THE DETERMINANTS OF BLOOD FLOW ON ECMO SUPPORT?

Said another way, what limits our ECMO blood flow? We will discuss how we can titrate ECMO blood flow in Chapter 17, but for now, let's focus only on the limits to blood flow. As we will discover, there are a host of limits to blood flow ranging from patient factors to factors associated with the circuit.

LIMITS OF THE MEMBRANE OXYGENATOR

Seems like with all of this technology and capability, the membrane oxygenator will be much better at oxygenation than our lungs, right? Actually, the oxygenator doesn't even come close to the potential capability of the lungs. Ultimately, it all comes down to the blood/gas interface. With the lungs, that interface comes in the form of the blood/alveolar interface. That means that the surface area of the 600 million alveoli in the lungs comes out to around 150 m^2, roughly the size of a tennis court. Compare this to the 4 m^2 surface area of the plastic fibers of most oxygenators, barely half of one of the service boxes. Additionally this interface is approximately 10–20 times thicker for the oxygenator than for the lungs, further hindering the diffusion of oxygen (Fig. 10.4).

Additionally, blood does not flow smoothly through the oxygenator, rather, it experiences a decrease in laminar flow both on the macro and micro levels as illustrated in Fig. 10.5.

As seen on the left, there will be areas throughout the oxygenator, such as right angles, corners, and any place where the flow of blood changes direction, where the flow of blood will slow down, giving rise to turbulence and ultimately decreasing the efficiency of the blood flow. The right illustrates how this can happen at the level of the plastic fibers, with blood changing direction to flow past the individual fibers.

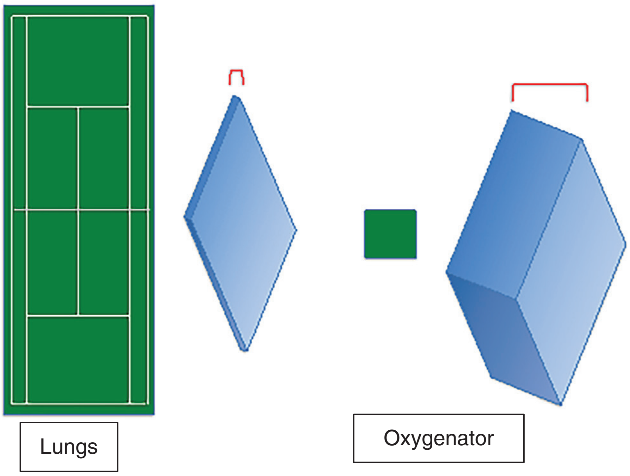

FIG. 10.4 Oxygenation capability of native lungs versus membrane oxygenator

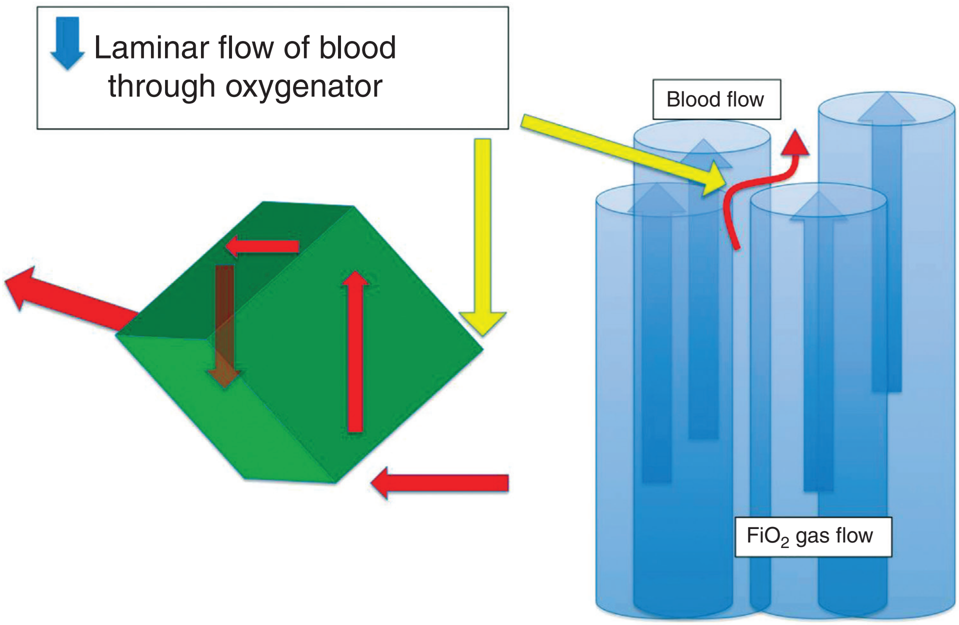

FIG. 10.5 Disruptions to laminar flow in membrane oxygenator

EFFECT OF MEMBRANE LIMITS: THE RATED FLOW OF THE MEMBRANE

With these inefficiencies and limits, we should be able to imagine how these add up to a maximum amount of flow that the membrane can tolerate, past which there would be no further increase in the content of blood (CaO_2) provided by the oxygenator (where $CaO_2 = 1.34 \times SaO_2 \times Hb + PaO_2 \times 0.003$) (Fig. 10.6).

Any further increase in blood flow past this maximum flow would just represent shunt, much in the same way that shunted blood in the lungs represents blood that shunts across without participating in gas exchange.

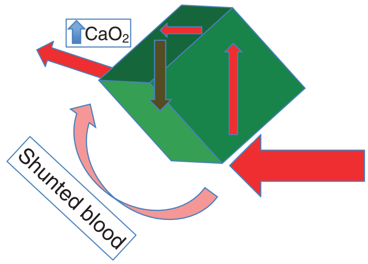

FIG. 10.6 Higher blood flow increases the proportion of shunted blood across the oxygenator

Let's now try to conceptualize what this looks like when it comes to how blood flows through an oxygenator. Imagine that you have an oxygenator with blood flowing through it with no limit on the amount of blood that can flow. If we were to plot the relationship between this blood flow and the oxygen content of the blood leaving the oxygenator, we would conceptualize something like what is shown in Fig. 10.7.

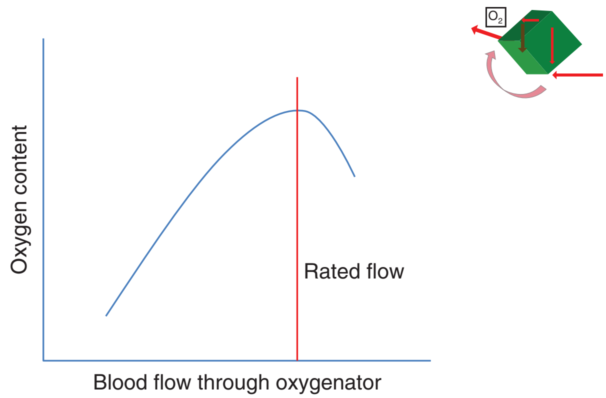

FIG. 10.7 The rated flow of the membrane oxygenator

In this graph, we see that at the beginning, as blood flow increases, the oxygen content would increase precipitously at first, then taper off as we approach the red line, due to shunting. However, as we continue to increase the blood flow, there reaches a critical point (red line), where the CaO_2 of blood leaving the oxygenator actually *decreases*, as the amount of shunted blood exceeds the non-shunted blood going through the oxygenator.

This point, where blood flow corresponds with the maximum content of oxygen that can be delivered, is referred to as the **rated flow** of the membrane.

Although every oxygenator that is being used clinically will perform differently depending on the age of the oxygenator, the level of anticoagulation, or the coagulation status of the blood, the rated flow, is a manufacturer-specified attribute, that is standard to all membrane oxygenators.

Since the rated flow represents the blood flow corresponding with the maximum CaO_2, it will usually be listed in the context of a standard hemoglobin and saturation change, with the convention that flow can raise the saturation of preoxygenator blood from 75% to 95% at a hemoglobin concentration of 12 g/dL.

BLOOD FLOW LIMITS OF THE CIRCUIT

So far, we have been discussing the limits of the membrane oxygenator with the assumption that there are no limits to blood flow in and out of the oxygenator. Now, let's consider the determinants of this blood flow and how these determinants drive and limit blood flow.

To further understand the dynamics of flow through a closed system, let's imagine two balloons filled with fluid, with Balloon 1 being completely filled up and Balloon 2 being empty, such that the pressure in Balloon 1 (P1) is much greater than the pressure in Balloon 2 (P2).

Now, let's connect these two balloons with some tubing. As you would imagine, there would be a flow of fluid in the direction of P1 to P2. What would determine the rate of flow? First, the pressure differential between the two balloons (P1 – P2) would play a large role. The more pressure the contents of Balloon 1 are under, the greater the flow of fluid (Fig. 10.8).

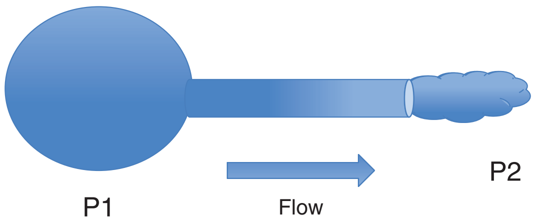

FIG. 10.8 Determinants of flow through a closed system

Past this point, the flow will be determined by the characteristics of the fluid and the tube connecting the two balloons. The thicker the fluid, the less flow you would anticipate. Additionally, the longer the tubing that connects the two balloons, the less flow, while the wider the tubing, the higher the flow.

This relationship of flow through a closed system is exactly what is happening in our ECMO circuit. Let's now apply these concepts to the circuit, drawing some conclusions that will be relevant to our clinical understanding and decision-making.

The standard equation for accounting for these variables when it comes to flow through a closed system is as follows:

$$\text{Flow} = \frac{\pi \times \Delta P \times r^4}{8 \times \text{viscosity} \times \text{length}}$$

Let's now put these variables into the context of the ECMO circuit. When we consider blood flow through an ECMO circuit, we have:

ΔP: pressure differential driven by the centrifugal pump and across circuit
r: width of cannulas and tubing (lesser extent)
length: length of tubing and cannulas (lesser extent)
viscosity: hematocrit and hemoconcentration

PRESSURE DIFFERENTIAL: ROLE OF CENTRIFUGAL PUMP

Remember in the example of our balloons, the greater the pressure differential, the greater the flow is. This is the essence behind the centrifugal pump that we introduced in Chapter 9.

To remind ourselves, the centrifugal pump consists of cones/fins that spin, creating a vortex, that *pulls* blood in through negative pressure and *pushes* the blood out through positive pressure (Fig. 10.9).

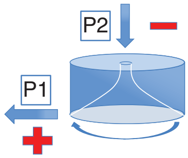

FIG. 10.9 The pressure differential generated by the centrifugal pump

The faster the pump spins, greater is the negative pressure generated (P2), and greater is the positive pressure generated (P1). Therefore, as the pump speed increases with higher revolutions per minute (RPMs), ΔP (P1 – P2) will also increase with a consequent increase in flow.

This increase in flow however, is limited by the **preload dependence** and **afterload sensitivity** of the centrifugal pump. Let's explore these concepts, which are fundamental to the centrifugal pump.

Effect of Preload Dependence on Pump Flow

Preload dependence is a distinct limit of the centrifugal pump. Just like the effect of preload on the cardiac myocytes introduced in Chapter 2, preload represents the filling pressure of the drainage system. This preload translates into the availability of blood for drainage necessary for the flow of blood through the pump.

This dependence on preload is even more profound for the centrifugal pump than for the heart due to the nature of the negative pressure generated by the pump. Even a modest drop in the pressure of blood in the venous drainage system (represented by central venous pressure [CVP]) can cause the pump flows to drop precipitously. Clinically, this means that often a drop in ECMO flows can precede a drop in mean arterial pressure (MAP) in the presence of hypovolemia.

There are two primary factors that determine the preload/filling pressures of the pump; the drainage cannula and the venous system from which the pump is draining blood.

Drainage Cannula

The drainage cannula can limit blood that can be pulled into the pump. The bigger the drainage cannula, the better the potential flow of blood. This is why the selection of the type/size/position of the drainage cannula is so important to the flow of blood and should always be considered when selecting cannulas at the beginning of an ECMO run. As you may recall from our discussion of cannula types, the typical drainage cannula is a multistage cannula, with multiple drainage holes throughout the length of the cannula that allow for increased flow relative to the negative pressure generated (Fig. 10.10).

Pressure of the Venous System

The number of drainage holes and size/position of the drainage cannula determine the negative pressure required to drain a specific volume of blood. This pressure is then exerted on the pressure of the venous system that the cannula is positioned in. If there is sufficient venous pressure, then the blood will be drained easily, facilitating blood flow into the pump.

However, if there is insufficient venous pressure (due to hypovolemia, a malpositioned cannula outside of the intrahepatic inferior vena cava (IVC), or increased abdominal pressures due to coughing as examples), then the negative pressure generated from the pump will be exerted onto

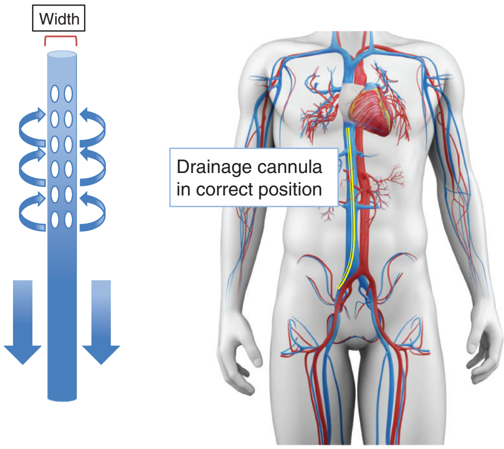

FIG. 10.10 Effect of drainage cannula size and position on pump filling pressures. (Modified from SciePro/Shutterstock.com.)

the vessel wall, causing it to suck down (position A to position B below), before returning to normal (position C) as flows drop (Fig. 10.11A–C).

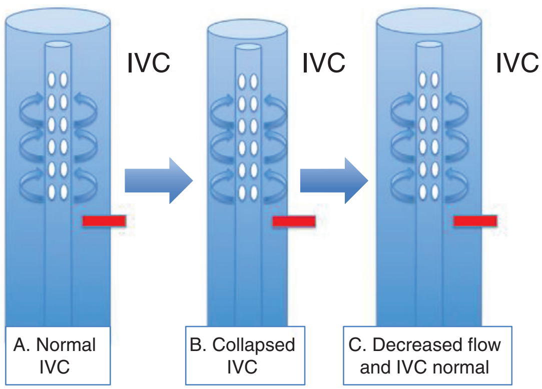

FIG. 10.11 The effect of insufficient venous pressure on the inferior vena cava
IVC, Inferior vena cava

The insufficient venous pressure causes a rapid drop in flow due to the unavailability of preload, which in turn decreases the negative pressure and releases the pressure exerted on the vessel wall. This process can repeat itself, causing the lines to jump and flows to fluctuate up and down, which is commonly referred to as "chatter."

The take home point is that preload is essential for the centrifugal pump to work. Without adequate filling pressures, the pump cannot generate a pressure differential, and flows will drop precipitously.

Effect of Afterload Sensitivity on Pump Flow

Let's now turn our attention to the positive pressure side of the pump. You will recall from our discussion of the pump in Chapter 9, we introduced the concept of afterload sensitivity, which stated that increased pressure downstream of the pump introduces afterload that the pump must push against.

As illustrated in Fig. 10.12, afterload decreases the gradient between P1 and P2.

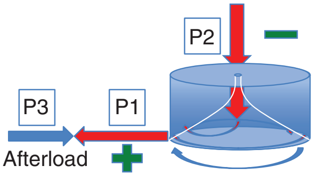

FIG. 10.12 The effect of afterload to the centrifugal pump on blood flow

The ultimate effect of this is a drop in ΔP, which will lead to an overall reduction in flow, as follows:

$$\Delta P = (P1 - P3) - P2$$

What Represents Afterload to the ECMO Pump?

Anything that increases pressure on the positive side of the pump can influence afterload. This includes the oxygenator, the return line, any additional source of increased resistance in this system (connectors/clots kinks/occlusions), and, ultimately, the pressure of the vascular system to which the blood is being returned.

If you are on VV ECMO, you are returning blood to the venous system and the afterload contribution of that system is the pressure of the venous system (~CVP). Contrast this to VA ECMO, where the afterload contribution of the vascular system is the pressure of the arterial system (~MAP), which you can anticipate to be much higher. Therefore, afterload sensitivity will be a bigger consideration for VA ECMO.

What Are the Clinical Implications of Afterload Sensitivity?

The pressure differential generated by the centrifugal pump always has to overcome this afterload, otherwise, retrograde flow will ensue. How can you mitigate this clinically when managing the patient, especially on VA ECMO?

1. When initiating ECMO, start with a fixed number of RPMs (usually around 1500) so that forward flow is maintained.
2. When weaning blood flows, maintain some RPMs so that you are evaluating lower flows, without experiencing retrograde flow.
3. If the pump stops flowing, clamp the circuit so that you will not have retrograde flow.

We will explore these more in the management section.

THE EFFECT OF RESISTANCE AND THE LIMITATION OF FLOW

We should now have a good sense of the effect of pressure differential (ΔP) on flow. Let's now turn towards the resistance of the system, and the effect that it has on overall flow. Resistance represents the other variables we identified in our system: radius, viscosity, and length as represented in our equation:

$$\text{Flow} = \frac{\pi \Delta P r^4}{8 \times \text{viscosity} \times \text{length}}$$

We will go over each of these and how they manifest in the ECMO circuit.

Radius and Limitations to Flow

Of the three variables, the radius of the system has the most significant effect on flow. That is why it is represented as r^4, signifying that any increase in the radius increases the flow by a factor to the fourth power. As an example, doubling the radius translates to a 16-fold increase in flow. Let's consider the effect of radius on flow as it relates to cannula size.

As an example, if you are trying to select between a 10-French cannula and a 25-French drainage cannula, you may be tempted to select the 19-French cannula, maybe to lessen the risk of venous obstruction, lower extremity compartment syndrome, or need for further dilation. You may rationalize that this only translates into a 2-mm change in diameter. However, this move actually translates into a three fold reduction in overall flow that the drainage cannula can accommodate! This highlights the importance of factoring in the role of radius of your circuit into the anticipated flow.

Length and Limitations to Flow

In a similar fashion, the longer the length of your system, the more resistance there is to flow. Let's return to our system connecting the two balloons. First imagine the flow between the balloons if the tubing connecting them was a foot. Now, imagine it is a mile long – you can envision the flow of fluid being significantly reduced, right (Fig. 10.13)?

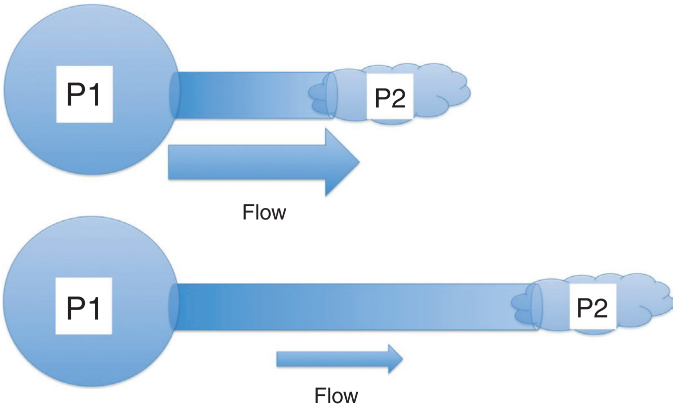

FIG. 10.13 The effect of length on flow

This becomes very relevant for the ECMO circuit. When building your circuit, you may come up with all kinds of reasons to build in more length – allow for greater mobility, facilitate transport, allow the patient to be pushed through the CT scanner, etc. Many of these are clinically essential. However, keep in mind the effect of length on overall flow.

Viscosity and Limitations to Flow

The primary determinant of viscosity is hematocrit, which can reflect red blood cells as well as hemoconcentration. For the most part, hematocrit/viscosity has a minimal effect on flow, as the level of hematocrit needed to have an appreciable effect on viscosity and ultimately flow becomes clinically untenable. If we were to plot the relationship of hematocrit to viscosity, we would have something similar to (Fig. 10.14), with a minimal effect on viscosity at levels seen in most patients, and an escalation in viscosity at levels of hemoglobin rarely seen in the critically ill patient population.

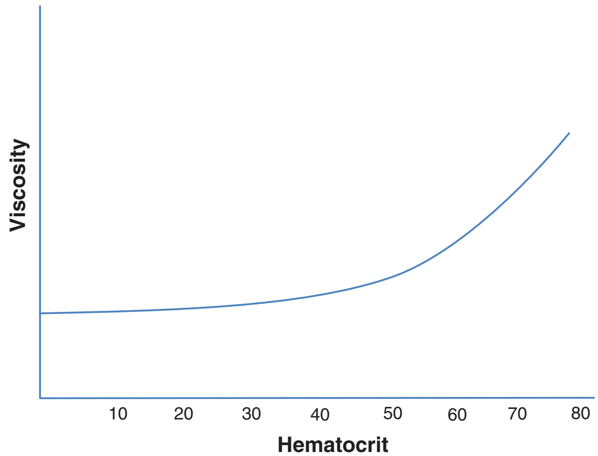

FIG. 10.14 The effect of hematocrit on viscosity

RPM Limits to Flow

We have now covered limits to flow as they relate to a theoretical closed system, but there are additional limits that are worth considering. The first is the limit of RPMs themselves. As a centrifugal pump spins at higher and higher speeds, you may not have any limitations to flow in terms of the rated flow of the membrane, preload dependence/afterload sensitivity of the pump, or resistance of the circuit.

However, at these high speeds, you may have a limitation due to the damaging effects of high RPMs on the blood that is being circulated through the pump. Specifically, high pump speeds can cause trauma to the red blood cells, manifesting as hemolysis. Besides the potential loss of blood cells, this hemolysis can cause a variety of adverse effects including renal toxicity, reduced immune function, potential for thrombosis, and alterations in hemodynamics.

Potential effects to monitor for hemolysis due to high pump speeds include jaundice, elevated plasma-free hemoglobin, elevated lactate dehydrogenase, elevated bilirubin, and dark urine. The presence of any of these may be a sign that RPMs are limiting or at the very least, should be limiting flow.

Effect of Recirculation Limiting Flow

One final effect related to the limits of flow that is unique to ECMO, specifically VV ECMO, is the concept of recirculation. Recirculation is an important concept in the management of VV ECMO that we will continue to discuss and review. In the current context, we will visit the implications of recirculation on the limits of flow.

Recirculation happens in VV ECMO, where blood is drained from the venous system and returned to the venous system. Ideally, deoxygenated blood is drained from the IVC, oxygenated, and then returned to the right atrium, where it can be returned to the body (Fig. 10.15).

Let's now zoom in and see what can happen as we increase blood flow (Fig. 10.16).

You can see that as we flow progressively higher, the drainage cannula exerts more of a negative pressure and eventually begins to pull oxygenated blood back into the circuit before it is circulated to the body. This has the effect of decreasing the effective blood flow. Any blood that is recirculated does not contribute to the oxygen delivery of the ECMO circuit, with the following relationship:

$$\text{Effective blood flow} = \text{ECMO blood flow} - \text{recirculation}$$

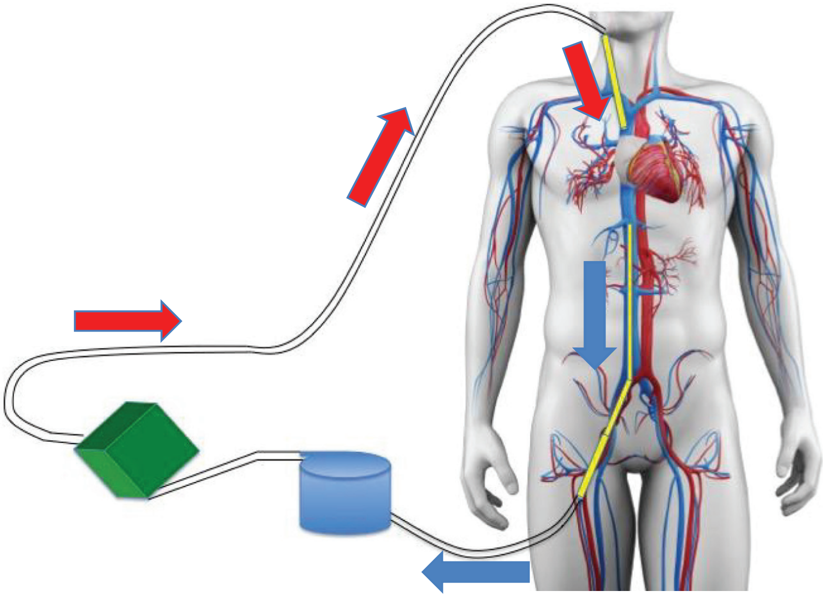

FIG. 10.15 Ideal ECMO blood delivery with drainage of deoxygenated blood and return of oxygenated blood. (Modified from SciePro/Shutterstock.com.)

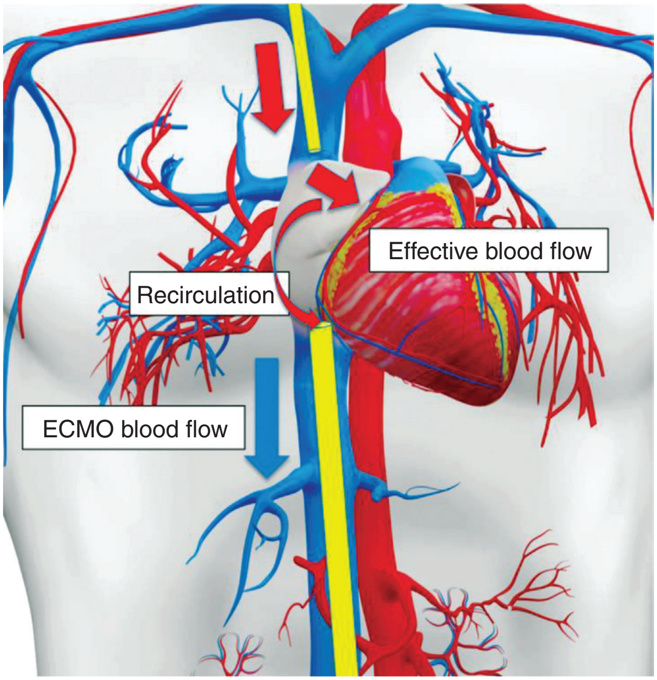

FIG. 10.16 The effect of recirculation on effective blood flow. (Modified from SciePro/Shutterstock.com.)

PUTTING IT ALL TOGETHER

Hopefully at the end of this chapter, you have a sense of flow, its role in oxygen delivery, and a grasp of the limitations to flow. There are many limitations to flow, and these should be carefully considered as flow is increased, decreased, or drops precipitously. Limits to flow play a large role in understanding how the ECMO blood flow is set. However, just because you are not limited in your flow does not necessarily mean that a certain flow should be maintained.

Rather, the titration of flow becomes a way to target the appropriate flow that matches both the capabilities of the circuit with the needs of the patient. We will continue to develop the tools needed to make this titration over the ensuing chapters, before diving more deeply into Flow Titration in Chapter 17. In the meantime, having a system to evaluate the circuit and assess its limitations like we have developed in this chapter is a significant step in the right direction.

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